

# JEE MAIN REVISION MASTER: PARABOLA SYLLABUS REDUCED

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 52 Analyzed PYQs

## 1. High-Yield Concept Priority Matrix

| Priority Tier | Concept Name                | Frequency | Category        |
|---------------|-----------------------------|-----------|-----------------|
| Very High     | Parabola - Core Application | 52 PYQs   | General Concept |

## 2. Master Formula Reference Card

| Concept / Topic | Mathematical Formulation & Governing Equations |
|-----------------|------------------------------------------------|
|-----------------|------------------------------------------------|

## 3. Deep-Dive Concepts & Linked PYQ Evidence

### 3.1 Parabola - Core Application (Very High Yield — 52 PYQs)

Core Principle: Key problem pattern in Parabola.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Let  $O$  be the vertex of the parabola  $y^2=4x$  and its chords  $OP$  and  $OQ$  are perpendicular to each other. If the locus of the mid-point of th..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "Let the parabola  $y=x^2+px+q$  passing through the point  $(1,-1)$  be such that the distance between its vertex and the  $x$ -axis is minimum..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Let  $O$  be the vertex of the parabola  $x^2=4y$  and  $Q$  be any point on it. Let the locus of the point  $P$ , which divides the line segment  $OQ$  in..." [Ans: N/A]

## 4. Benchmark Illustrative PYQs with Master Solutions

**Benchmark Problem 1: JEE Main 2026 (Online) 8th April Evening Shift (Q#1)**

Let O be the vertex of the parabola  $y^2=4x$  and its chords OP and OQ are perpendicular to each other. If the locus of the mid-point of the line segment PQ is a conic C, then the length of its latus rectum is :

- (A) 1
- (B) 2
- (C) 4
- (D) 8

**Correct Answer: None**

**Solution Analysis:**  $t_1 t_2 = -4$   $2\lambda = t_1^2 + t_2^2$   $2k = 2(t_1 + t_2)$   $2\lambda = k^2 - 2t_1 t_2$   $2\lambda = k^2 + 8$   $k^2 = 2\lambda - 8 = 2(\lambda - 4)$   
L.R.=2